Discrete Math HW6

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1. $\begin{pmatrix}0 & 0 & 1\\1 & 1 & 2\\2 & 2 & 3\\3 & 3 & 4\end{pmatrix} \cdot \begin{pmatrix}-1 & -1\\2 & 2\\1 & 1\end{pmatrix} \cdot \begin{pmatrix}4\\1\end{pmatrix}$

$$\begin{pmatrix}0 & 0 & 1\\1 & 1 & 2\\2 & 2 & 3\\3 & 3 & 4\end{pmatrix} \cdot \begin{pmatrix}-1 & -1\\2 & 2\\1 & 1\end{pmatrix} \cdot \begin{pmatrix}4\\1\end{pmatrix} = \begin{pmatrix}1 & 1\\3 & 3\\5 & 5\\7 & 7\end{pmatrix} \cdot \begin{pmatrix}4\\1\end{pmatrix} = \begin{pmatrix}5\\15\\25\\35\end{pmatrix}$$

2. F(x)=$3x^2 - 2x + 5$ $F(A) = ?$

$$A = \begin{pmatrix}1 & -2 & 3\\2 & -4 & 1\\3 & -2 & 5\end{pmatrix}$$

Considering addition of scalar c to matrix A as A+c $\rightarrow A + Ic$:

$A^2 = \begin{pmatrix}6 & 0 & 16\\-3 & 10 & 7\\14 & -8 & 32\end{pmatrix}$

$3A^2 = \begin{pmatrix}18 & 0 & 48\\-9 & 30 & 21\\42 & -24 & 96\end{pmatrix}$

$2A = \begin{pmatrix}2 & -4 & 6\\4 & -8 & 2\\6 & -4 & 10\end{pmatrix}$

$$3A^2 - 2A = \begin{pmatrix} 16 & 4 & 42 \\ -13 & 38 & 19 \\ 36 & -20 & 86 \end{pmatrix}$$

$$F(A) = 3A^2 - 2A + 5I = \begin{pmatrix} 21 & 4 & 42 \\ -13 & 43 & 19 \\ 36 & -20 & 91 \end{pmatrix}$$

3. Find all $B_{[2X2]}$, such that $\mathrm{B}^2 = 0$.

   Consider $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

   Therefore $B^2 = \begin{pmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{pmatrix} = 0$

   Therefore,

   $$\begin{cases} a^2 + bc = 0 \\ ab + bd = 0 \\ ac + cd = 0 \\ bc + d^2 = 0 \end{cases} \rightarrow \begin{cases} a^2 = -bc \\ b(a+d) = 0 \\ c(a+d) = 0 \\ d^2 = -bc \end{cases} \rightarrow \begin{cases} a + d = 0 \\ bc = -a^2 \end{cases}$$

   Therefore, for all matrices $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, such that $a + d = 0$ and $bc = -a^2$, expression $B^2 = 0$ holds.

4. $\mathrm{A} = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 4 & -1 & 5 & -1 \end{pmatrix}$. Find $\mathrm{A}^T,\ A \cdot A^T,\ A^T \cdot A$

   $$A^T = \begin{pmatrix} 1 & 4 \\ 2 & -1 \\ 1 & 5 \\ 3 & -1 \end{pmatrix}$$

   $$A * A^T = \begin{pmatrix} 15 & 4 \\ 4 & 43 \end{pmatrix}$$

   $$A^T * A = \begin{pmatrix} 17 & -2 & 21 & -1 \\ -2 & 5 & -3 & 7 \\ 21 & -3 & 26 & -2 \\ -1 & 7 & -2 & 10 \end{pmatrix}$$

5. Find: $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$

Following cofactor expansion method:

$det(M) = det\begin{pmatrix} 5 & 6 \\ 8 & 9 \end{pmatrix} - 2 * det\begin{pmatrix} 4 & 6 \\ 7 & 9 \end{pmatrix} + 3 * det\begin{pmatrix} 4 & 5 \\ 7 & 8 \end{pmatrix} = (45 - 48) - 2(36 - 42) + 3(32 - 35) = -3 + 12 - 9 = 0$

6. Find x

$$\begin{vmatrix} 3 & x & -x \\ 2 & -1 & 3 \\ x+10 & 1 & 1 \end{vmatrix} = 0$$

Following cofactor expansion method:

$det\begin{pmatrix} 3 & x & -x \\ 2 & -1 & 3 \\ x+10 & 1 & 1 \end{pmatrix} = 3det\begin{pmatrix} -1 & 3 \\ 1 & 1 \end{pmatrix} - x * det\begin{pmatrix} 2 & -1 \\ x+10 & 1 \end{pmatrix} + x * det\begin{pmatrix} 2 & -1 \\ x+10 & 1 \end{pmatrix} = 3(-1-3) - x(2-3x-30) - x(2+x+10) = -12 + 3x^2 + 28x - 12x + x^2 = x^2 + 8x - 6 = 0$

$x_{1,2} = \frac{-8 \pm 2\sqrt{22}}{2} = -4 \pm \sqrt{22}$

Thus, $x_1 = -4 + \sqrt{22}, x_2 = -4 - \sqrt{22}$